

Differential spatial regulation and activation of integrin nanoclusters inside focal adhesions

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In this manuscript, the authors analyse the nanoscale localisation of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins in integrin adhesion complexes (IAC) by dual-colour STORM and assess the spatial organisation at the nano and mesoscale of their main adaptors (paxillin, talin and vinculin). This is an **important** work that provides detailed analyses that reveal how elements of these complex structures are really organised at the nanoscale, an essential perspective for a better understanding of how IACs function and regulate mechanotransduction processes. The evidence presented is **solid**, with super-resolution imaging experiments conducted using a single, validated methodology and subsequent computational modelling that enabled a quantitative assessment of the resulting data.

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Abstract

$\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins are core components of focal adhesions (FAs) involved in cell attachment, migration and mechanobiology-dependent processes. Recent works indicate that both integrins organize in nanoclusters inside FAs, with sub-populations of active and inactive β_1 nanoclusters. However, whether both integrins work in concert or their activities are spatially regulated is not fully understood. Using dual-color super-resolution microscopy we show that integrins $\alpha_5\beta_1$ and $\alpha_v\beta_3$ exhibit similar functional nanoscale segregation. Notably, both integrins never mix at the nanoscale indicating that their functions might be spatially regulated. We find a nearly 1:1 stoichiometry between active integrin and adaptor nanoclusters suggesting that coordinated integrin activation occurs via the concurrent engagement of adaptor nanoclusters. Interestingly, $\alpha_5\beta_1$ nanoclusters preferentially localize at the FA periphery in close proximity to adaptor nanoclusters, establishing regions of multi-nanocluster enrichment, whereas $\alpha_v\beta_3$ nanoclusters uniformly distribute throughout FAs. Overall, our results show that adhesion proteins arrange as modular nanoscale units that distinctively organize inside FAs to spatially regulate integrin activation and function.

Introduction

Adhesion complexes (ACs) in adherent cells, such as focal adhesions (FAs) and fibrillar adhesions (fAs) are places for cell attachment, mechanosensing and mechanotransduction (1–4). These structures are composed by different integrin receptors that bind to their ligands on the extracellular matrix (ECM), and by a myriad of intracellular adaptors, proteins that dynamically connect integrins to the actin cytoskeleton machinery relaying the transmission of mechanical forces in a bidirectional manner (5–7). Studies over the last two decades have provided a wealth of information focused on understanding these complexes, the proteins they contain, their role in cellular processes, such as mechanosensing, migration and proliferation, as well as how influential their regulation is in developmental and pathogenic processes (8–15).

Integrins $\alpha_5\beta_1$ and $\alpha_v\beta_3$ are transmembrane heterodimeric proteins critical for the formation of ACs. Both $\alpha_5\beta_1$ and $\alpha_v\beta_3$ bind to the RGD peptide motif of certain ECM components, such as fibronectin (16–18), and are linked to the cell's cytoskeleton via many of the same adaptor proteins (19–22). The apparent redundancy of different integrins binding to the same ligand and same adaptor proteins is intriguing and has been the subject of intense research over the past years. Earlier studies on primary fibroblasts showed that these integrins segregate in different adhesion structures, indicating different roles, with $\alpha_5\beta_1$ being enriched in fAs (23), adhesive sites at central regions of the cell, and $\alpha_v\beta_3$ being confined to focal complexes at the base of cellular protrusions, and to FAs, which anchor contractile stress fibers at the periphery of the cell (24). However, more recent super-resolution microscopy studies showed that both types of integrins are simultaneously present inside FAs (25). Within FAs, $\alpha_5\beta_1$ has been proposed as the integrin responsible for adhesion strength while $\alpha_v\beta_3$ enables mechanotransduction, as cells exposed to $\alpha_5\beta_1$ blocking antibody exhibited weaker attachment than those whose $\alpha_v\beta_3$ integrins were blocked (5). These results are also aligned to *in vitro* reconstitution experiments, that have assessed the catch-bond behavior of both integrins with fibronectin. In this type of bond, the lifetime of the bond increases as the applied force increases, up to a certain threshold, beyond which the bond ruptures. The rupture force for $\alpha_5\beta_1$ binding to its ligand has been found to be higher (~10–30 pN) than that of $\alpha_v\beta_3$ (~5–25 pN) (26, 27). Nevertheless, single molecule tracking data indicated that $\alpha_v\beta_3$ integrins are mainly stationary inside FAs and more stably bound to the ligand, favoring adhesion as its main role (25). It has been also shown that $\alpha_5\beta_1$ is involved in Rho activation during late spreading, enhancing RhoA-mediated actomyosin contractility (24, 28), supporting the idea that $\alpha_5\beta_1$ is involved in force-mediated adhesion reinforcement. In a *tour-de-force* effort, Fässler and colleagues found that only the simultaneous expression of both integrins enabled cells to sense the rigidity of the ECM and to respond by reinforcing the actomyosin structure (29). These results, supported by others (30), have led to the recent notion that both integrins work in a synergistic manner, exhibiting cooperativity during cell adhesion, migration and mechanosensing, although the mechanisms are far from being fully resolved (30).

The highly dynamic conformational states exhibited by integrins together with their lateral and axial nanoscale organization might play important roles regulating the differential or synergistic functions of different integrins inside FAs. Indeed, it has been shown that integrins can rapidly transition between different conformational states that relate to their activation and binding capacity to their ligands (25, 31, 32). Moreover, in addition to the nanoscale axial segregation of different AC components inside FAs (33), it has been reported that integrins as well as different adaptor proteins also segregate laterally into small nanoclusters inside FAs (34–37). Recent super-resolution studies have further shown that β_1 integrins segregate in both active and inactive nanoclusters inside FAs (36) and single-molecule tension measurements uncovered distinct sub-populations of load bearing integrins and a high fraction of unengaged integrins (38, 39). Together, these results suggest that at any given time only a

fraction of integrins might be active inside FAs and engaged with their adaptor partners. Such a dynamic scenario would allow integrins to differentially interact with their environment and possibly diversify their functions. As $\alpha_5\beta_1$ and $\alpha_v\beta_3$ have different activation states and are both contained in FAs, it is plausible that their spatial regulation and activation state inside FAs provides a physical mechanism to tune their functions. However, such studies have not been reported yet.

Here, we combined dual-color super-resolution microscopy together with robust analytical tools to study a number of endogenous FA proteins on human fibroblast cells. Specifically, we imaged one of the two integrins ($\alpha_5\beta_1/\alpha_v\beta_3$), their full population or active conformations, with each of three key adaptor proteins, paxillin, talin and vinculin. Our results show that each of these proteins form segregated nanoclusters of remarkably similar size and molecular density inside FAs, and interestingly, also on the membrane outside of ACs. Moreover, we find sub-populations of active and inactive β_1 and β_3 nanoclusters inside FAs, which spatially segregate from each other and provide evidence for a nearly 1:1 stoichiometry between talin, vinculin and active integrin nanoclusters. Interestingly, we find that $\alpha_v\beta_3$ nanoclusters organize randomly inside FAs regardless of their activation state, whereas $\alpha_5\beta_1$ and in particular the active conformation of β_1 nanoclusters preferentially organize at the edges of FAs. Overall, we propose that the nano- and meso-scale organization of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins and their distinct lateral segregation and activation state are key factors in the differentiation of their roles within FAs.

Results

Proteins of the ACs organize as universal nanoclusters across regions of the basal membrane

We employed dual-color super-resolution microscopy based on stochastic optical reconstruction microscopy (STORM) to determine how adhesion proteins (integrins and key adaptor proteins) are spatially organized at the nanoscale. Human fibroblasts (HFF-1 cells) were seeded on fibronectin (FN)-coated glass slides for 24 h, after which the cells were processed for immunofluorescence. One integrin (either the total population of $\alpha_5\beta_1$ or $\alpha_v\beta_3$, or the active population specific to the β_1 or β_3 integrin subunit) was labeled together with one adaptor protein (paxillin, C terminus of talin-1, or vinculin) and then imaged using STORM. We note that while the pan antibodies used to target the full population of integrins are specific to $\alpha_5\beta_1$ or $\alpha_v\beta_3$, (with the anti- α_5 antibody binding the α_5 subunit, which only dimerizes with β_1 , and the anti- $\alpha_v\beta_3$ targeting this specific heterodimer), the active antibodies target the β subunits. These β subunits can dimerize with other α subunits, but with our experimental conditions, where FN is the most abundant ligand (see Materials & Methods), the primary integrins being probed are the heterodimers, $\alpha_5\beta_1$ or $\alpha_v\beta_3$. Given the huge number of adaptor proteins that have been reported to interact with integrins (19, 20), we chose for paxillin, talin and vinculin since they belong to the core-components of the adhesome (21) and crucially locate at distinct axial layers of functional activity, i.e., paxillin at the signaling layer, while talin and vinculin locate at the force-transduction layer (33).

Super-resolution images qualitatively revealed an inhomogeneous distribution for all the proteins examined, forming small and laterally segregated nanoclusters inside FAs (Fig. 1A–D). To quantitatively substantiate these observations, we first used a mask to classify all the single molecule localization events according to their location on the membrane, i.e., FAs, fAs or outside ACs (see Materials & Methods, Fig. S1A–F). To group localizations into nanoclusters we used a density-based spatial clustering of applications with noise (DBSCAN) algorithm (40), which has been extensively used for quantifying protein nanoclustering from single molecule localization images, including β_1 integrins inside FAs (36, 41). Since STORM data consist of lists of single-molecule localizations and each labeled protein can be localized multiple times due to fluorophore blinking (42), it is important to first assess the number of localizations detected due to the

multiple blinking events of a single fluorophore. For this, we followed an approach similar to that reported by Spiess et al. to generate distributions of the number of localizations obtained from isolated fluorescent spots on glass next to the cells, for both channels, these will mostly correspond to individual antibodies (Abs) (**Fig. 1E**, Materials & Methods). By doing this approach one can determine the number of localizations corresponding to individual Abs and infer whether the clustered localizations observed correspond to individual or multiple proteins. The median values of the distributions were 5 localizations for Alexa 405-Alexa 647-conjugated Abs (Channel 1) and 3 localizations for Cy3-Alexa 647-conjugated Abs (Channel 2). Moreover, to further minimize the effect of fluorophore blinking on cluster identification, we imposed a threshold of 10 localizations as commonly used in the field (36, 41). Thus, any cluster identified by DBSCAN with 10 or more localizations was considered as a cluster of more than one protein (see Materials & Methods). DBSCAN classified between 70% to 80% of the localizations of each of the proteins as belonging to nanoclusters, regardless of whether they were located inside FAs or on regions of the membrane outside ACs (Fig. S1G-I). Notably, the number of localizations per nanocluster and their size were similar across the different adhesion proteins (integrins and adaptors) inside FAs, with a median of ~20 localizations per nanocluster, likely corresponding to a few copies of the protein, and a diameter of ~40–45 nm (**Fig. 1F, G**).

Similar results in terms of size and number of localizations per nanocluster were obtained when performing the analysis outside ACs (Fig. S2). In addition, the nanocluster characteristics were maintained when specifically labeling the active integrin (β_1 or β_3) populations (**Fig. 1D, F, G**). These results are in good quantitative agreement with previous reports using STORM showing that both active and inactive integrin β_1 nanoclusters have a median size of 40 nm and 35 nm, respectively, as well as comparable number of single molecule localizations per nanocluster (36). Overall, our results agree with previous work on $\alpha_5\beta_1$ and provides new insight on the integrin $\alpha_v\beta_3$ and the main adaptor proteins, paxillin, talin and vinculin, indicating that organization into laterally segregated nanoclusters of similar size and number of molecules is maintained across different protein types. Interestingly, such nanoscale organization is independent of whether these adhesion proteins partition inside FAs or outside ACs (Fig. S2).

Global mesoscale organization of adhesion protein nanoclusters inside FAs

We next sought to investigate how the different adhesion proteins laterally organize inside FAs beyond their nanoclustering, i.e., at the mesoscale. For this, we first quantified the density of nanoclusters per unit area of the membrane, within FAs (**Fig. 2A**) and outside ACs (**Fig. 2B**). As expected, the density of nanoclusters was larger in FAs as compared to outside adhesions (~5–10-fold increase) (**Fig. 2A, B**). Adaptor proteins, such as talin or vinculin, are shared between $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins (21), therefore we compared the density of the full population of integrin nanoclusters (~36 μm^{-2} in FAs) to the density of adaptor protein nanoclusters (~10–12 μm^{-2} for talin and vinculin, ~20 μm^{-2} for paxillin in FAs). These data reveal a significantly higher (~2–4 fold) density of integrin nanoclusters in comparison to their adaptors and, consequently, a stoichiometric nanocluster mismatch between the integrins and the adaptors, regardless of whether they partition inside or outside FAs (**Fig. 2A, B**). Since the active integrin conformation is the one engaged with the adaptors (43, 44), we also quantified the density of active integrin nanoclusters (active β_1 and active β_3) per unit area inside FAs. Our results revealed that only ~30% of the total $\alpha_5\beta_1$ nanocluster population were in an active conformation, and this percentage was even lower for β_3 nanoclusters (~20%) (**Fig. 2A**), indicating that at any given time there is a large percentage of inactive integrin nanoclusters inside FAs. We note that these percentages might be even lower considering that our Abs for mapping the active conformations are not exclusive for $\alpha_5\beta_1$ and $\alpha_v\beta_3$, but rather to the total active β_1 and β_3 sub-populations. Notably, the total density of active β_1 and β_3 integrin nanoclusters stoichiometrically matches that of talin and vinculin nanoclusters, and it is only slightly outnumbered by paxillin nanoclusters inside FAs (**Fig. 2A**), suggesting that these active integrins might be fully engaged to their adaptors. The modest

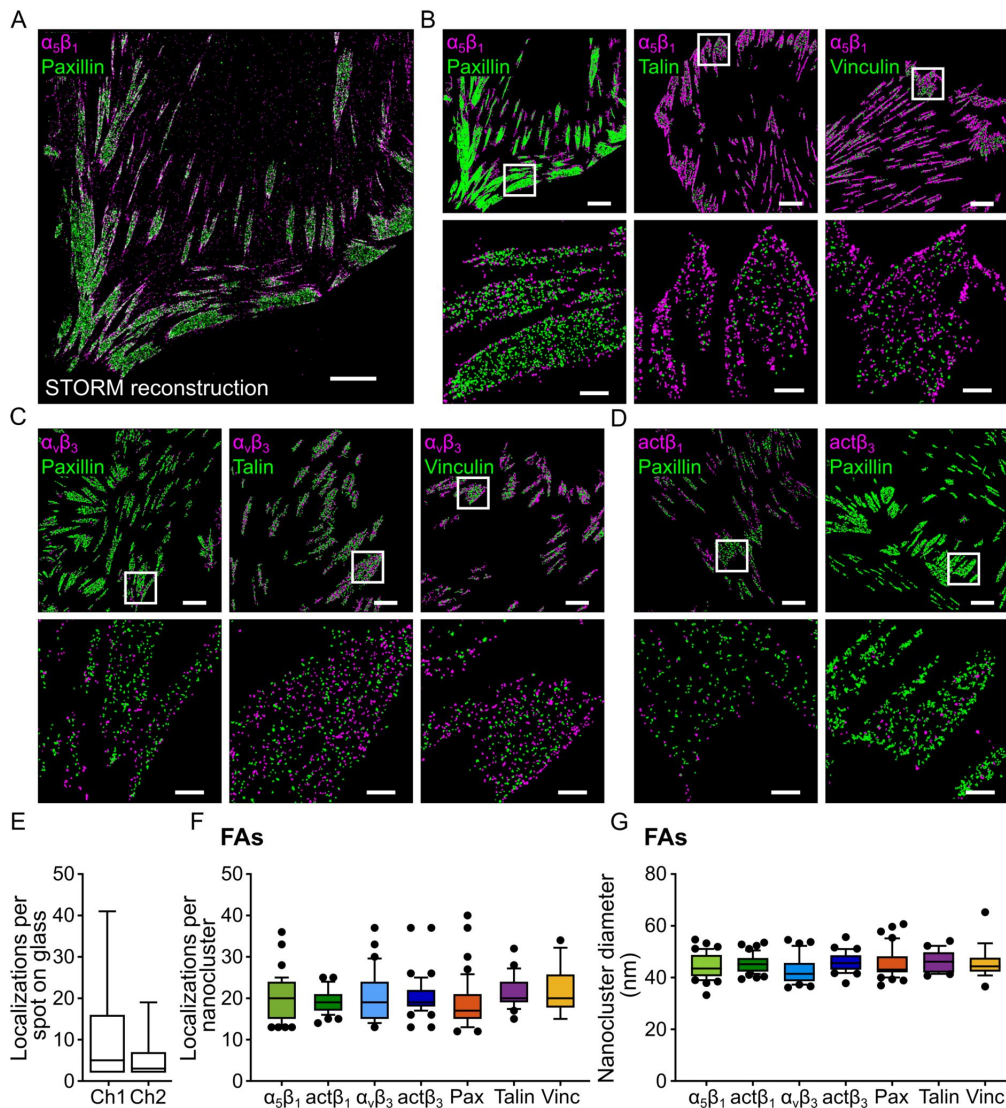


Figure 1

Proteins of FAs organize as universal nanoclusters of similar size and containing a similar number of proteins.

(A) Representative dual-color STORM image of an HFF-1 cell fixed after 24 h on FN and labeled for $\alpha_5\beta_1$ (magenta) and paxillin (green). Scale bar: 5 μm . (B) Representative images of all the adhesions identified for $\alpha_5\beta_1$ with paxillin (left), talin (middle) and vinculin (right), with zoomed-in regions shown below each panel. Scale bar top row: 5 μm , bottom row: 1 μm . (C) Representative images of all the adhesions identified for $\alpha_v\beta_3$ with paxillin (left), talin (middle) and vinculin (right), with zoomed-in regions shown below each panel. Scale bar top row: 5 μm , bottom row: 1 μm . (D) Representative images all the adhesions identified for of active β_1 (act β_1) with paxillin (left), active β_3 (act β_3) with paxillin (right), with zoomed-in regions shown below each panel. Scale bar top row: 5 μm , bottom row: 1 μm . (E) Distribution for the number of localizations per secondary antibody on glass next to cell regions for both of the secondary antibodies used in this study, goat anti-mouse Alexa 405-Alexa 647 (Ch1) and goat anti-rabbit Cy3-Alexa 647 (Ch2). The median number of localizations per spot corresponds to: 5 for Alexa 405-Alexa 647-conjugated antibodies and 3 for Cy3-Alexa 647-conjugated antibodies (see Materials & Methods). (F, G) Box-and-whisker plots showing the number of localizations per nanocluster (F) and the nanocluster diameter (G) for the proteins in this study. The box represents the range from the 25th to the 75th percentiles and whiskers from 10th to 90th percentiles, with the median indicated by a horizontal line. Points outside these whiskers correspond to outliers. Individual points (points within the whiskers not shown) correspond to the median value over all nanoclusters for each cell (for N values, i.e., number of independent experiments, see SI Table S5). No statistical differences were found using the multi comparison one-way ANOVA test, where not significant (ns) corresponds to $p > 0.05$.

percentage of active β_1 integrin nanoclusters observed in our data ($\sim 30\%$, **Fig. 2A**) is in close agreement with a recent report from Spiess *et al.*, obtained using a different super-resolution technique (36) and consistent with the high percentage of low-force bearing integrins measured by using single-molecule force sensors (39). In addition to the $\alpha_5\beta_1$ results, our data also reveal the existence of a sub-population of active β_3 nanoclusters inside FAs (**Fig. 2A**). Together, these results strengthen the hypothesis proposed by Spiess *et al.* of a coordinated mechanism that laterally segregates active and inactive nanoclusters inside FAs (36).

Interestingly, despite this common feature, we found a significantly higher density of $\alpha_5\beta_1$ nanoclusters as compared to $\alpha_v\beta_3$ (**Fig. 2A, B**). Moreover, the percentage of active β_1 integrin nanoclusters was larger than for active β_3 , in particular inside FAs (30% and 20% for active β_1 and active β_3 , respectively) (**Fig. 2A**). To further enquire whether both types of integrin nanoclusters act in concert by associating at the nanoscale and sharing their vinculin and talin partners, we performed dual color STORM imaging of the total populations of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins (**Fig. 2C**). For this, we used a rabbit anti- α_5 antibody (see SI, Table S2) to enable simultaneous labeling with the mouse anti- $\alpha_v\beta_3$ antibody that we used throughout this study. Visual inspection as well as quantification of the images show that $\alpha_5\beta_1$ and $\alpha_v\beta_3$ do not intermix at the nanoscale, forming distinct nanoclusters that spatially segregate from each other inside FAs (**Fig. S3A, B**). The differences in expression levels, activation states, and lateral nano- and meso-scale segregation of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ suggest the existence of a mechanism that defines the distinct functional roles of these two integrins inside FAs.

To get a better understanding on the global organization of both types of integrins in relation to their main adaptor proteins, we calculated the edge-to-edge distance between the nearest-neighbor integrin nanoclusters and each of the adaptors (see Materials & Methods). Although visual inspection of the dual color STORM images already suggested that integrin nanoclusters are also physically segregated from their adaptor partners (**Fig. 1B, C**) we used this quantitative approach to estimate the degree of colocalization (i.e., spatial overlap) between different nanoclusters. Since we measure edge-to-edge distances between nearest nanoclusters (EED-NN), values below zero correspond to physically overlapping nanoclusters, EED-NN values above zero to spatially segregated nanoclusters, and EED-NN = 0 corresponds to contiguous nanoclusters, i.e., next to each other. Our results quantitatively confirm that the large majority of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ nanoclusters are laterally segregated from their adaptor partners (**Fig. 2D**, **Fig. S3C, D**). Their spatial segregation is essentially determined by the density of nanoclusters within the FAs, as similar EED-NN distributions were obtained from simulations of particles uniformly distributed inside the experimentally obtained FAs, using identical particle densities as to the ones obtained experimentally for the integrin nanoclusters and the different adaptors (Materials & Methods) (**Fig. 2D**, black lines). Considering the canonical view that integrins inside FAs are bound to paxillin and engaged to talin and vinculin (43, 45), these results might be surprising at first sight, as one would expect a much higher fraction of EED-NN values below zero. Nevertheless, using different super-resolution microscopy techniques, others have also found negligible spatial colocalization of integrins with these adaptor proteins (36, 37). Moreover, Dunn and co-workers also found that most ligand-bound integrins inside FAs exist in a minimally tensioned state (< 2 pN) and not dependent on the actin cytoskeleton (39). The results from the Dunn lab are fully in line with the lack of colocalization of integrins to talin and vinculin as we determined here, since these two adaptors mediate the linkage of integrins to the actin cytoskeleton (43, 46). Nevertheless, we would like to mention that we labeled talin at its actin-binding C-terminal region, which in a 2D projection can appear to be at a distance of 100 nm or more from the N-terminal integrin-binding site (47), so lack of spatial overlap does not necessarily indicate lack of direct engagement.

It has been recently proposed that adhesion proteins including integrins organize in a hierarchical fashion inside FAs, forming nanoclusters similar to the ones reported here, that further enrich into compartmentalized regions around 300 nm in size (37). To enquire whether such hierarchical

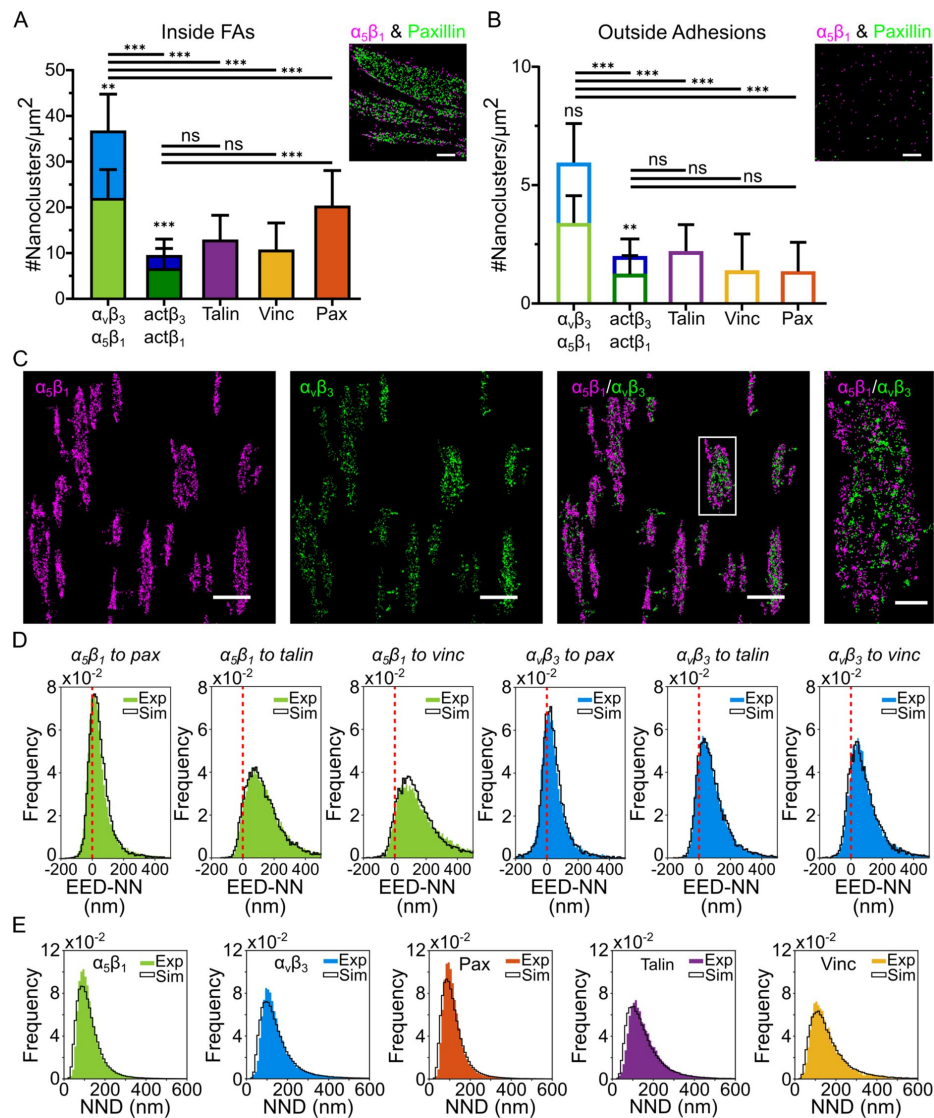


Figure 2

Mesoscale organization of integrin nanoclusters and adaptor proteins inside FAs.

(A, B) Mean density of nanoclusters per unit area inside FAs (A) and on the cell membrane outside adhesions (B), with $\alpha_5\beta_1$ and $\alpha_v\beta_3$ in light and dark green, respectively; and $\alpha_v\beta_3$ and $\alpha_5\beta_1$ in light and dark blue, respectively. The densities of integrins nanoclusters per unit area are shown as stacked graphs to allow comparison between the total number of integrin nanoclusters ($\alpha_5\beta_1$ and $\alpha_v\beta_3$) and the adaptor proteins. The bars show the mean $\pm$ st. dev of the distribution. Statistical significance between data of all proteins was tested using a multi comparison one-way ANOVA, assuming Gaussian distribution of points. The comparison between $\alpha_5\beta_1$ and $\alpha_v\beta_3$, ($\alpha_5\beta_1$ and $\alpha_v\beta_3$) was done with an unpaired t-test, where ns = not significant, $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. The insets show representative images of nanoclusters for $\alpha_5\beta_1$ (magenta) and paxillin (green) in FAs (A) and outside adhesions (B). Scale bar: 1 μ m. (C) Representative dual-color STORM image of the total populations of $\alpha_5\beta_1$ (magenta) and $\alpha_v\beta_3$ (green) integrins inside FAs. The panel on the far-right side corresponds to a zoomed-in region of a single FA (white box) showing spatially segregated $\alpha_5\beta_1$ and $\alpha_v\beta_3$ nanoclusters. Scale bars: 2 μ m, and 500 nm for the zoomed-in region. (D) Histogram of the EED-NN values between $\alpha_5\beta_1$ and the different adaptors (green) and, between $\alpha_v\beta_3$ and the adaptors (blue). The black line in each plot corresponds to the histogram for the simulated data sets. The dashed red line at zero corresponds to nanoclusters contiguous to each other. Percentage of overlapping nanoclusters: $\alpha_5\beta_1$ -paxillin: exp = 28.4%, sim = 24.4%; $\alpha_5\beta_1$ -talin: exp = 6.7%, sim = 8.1%; $\alpha_5\beta_1$ -vinculin: exp = 6.0%, sim = 7.7%; $\alpha_v\beta_3$ -paxillin: exp = 31.7%, sim = 29.2%; $\alpha_v\beta_3$ -talin: exp = 15.7%, sim = 16.5%; $\alpha_v\beta_3$ -vinculin: exp = 14.9%, sim = 17.4%. (E) Histogram of the NND values between nanoclusters of the same protein. The black line in each plot corresponds to the histogram for the simulated data sets. Bin width is 10 nm.

organization is present in our data, we performed nearest-neighbor distance (NND) analysis between nanoclusters of the same proteins (**Fig. 2E**). In addition, we implemented simulations of uniformly distributed particles inside the experimental FA regions, using the same experimental nanoclusters densities, and made the corresponding NND analysis (**Fig. 2E**). Notably, our results show that the global organization of all the adhesion protein nanoclusters investigated is highly uniform inside FAs, as the experimental NND distributions were indistinguishable from those of randomly distributed particles (**Fig. 2E**). In contrast to the recent findings of the Kusumi lab (37), we did not observe any evidence of spatial enrichment of protein nanoclusters inside FAs. Instead, the observed characteristic separation between nanoclusters appears to be solely determined by their density inside FAs.

Integrin $\alpha_5\beta_1$ but not $\alpha_v\beta_3$ nanoclusters are enriched on FA edges

Our previous global analysis rendered a uniform inter-cluster organization of all adhesion protein nanoclusters inside FAs. Yet, visual inspection of the dual-color STORM images suggests that a significant fraction of $\alpha_5\beta_1$ nanoclusters preferentially localize on FA edges (**Fig. 1B**). To quantify these observations, we calculated the shortest-path distance from each $\alpha_5\beta_1$ nanocluster to the edge of its corresponding FA, i.e., its distance-to-FA-edge, and generated a histogram of these values (**Fig. 3A**). In addition, we generated a similar histogram from simulations of uniformly distributed $\alpha_5\beta_1$ nanoclusters on FAs (see Materials & Methods). In general, if the nanoclusters were uniformly distributed throughout the FA, one would expect that the probability of finding a nanocluster at a specific distance to the FA edge decreases with increasing distance. This occurs because there is a larger area along the perimeter of any given shape and thus a higher number of nanoclusters (e.g., for a circular shape, the decrease is linear with the distance to the FA edge). We found that our experimental and simulated data indeed exhibit this general trend, although, notably, a higher percentage of $\alpha_5\beta_1$ nanoclusters were detected at distances below 100 nm from the FA edges, as compared to a uniform distribution (**Fig. 3A** and *inset*), indicative of their preference to locate closer to the FA edges. Similar trends were also obtained for talin, and to a lesser extent for paxillin, whereas the distributions of $\alpha_v\beta_3$ and vinculin were indistinguishable from random (**Fig. S4A**).

To better assess the extent of nanocluster enrichment at FA edges at the single cell level, we computed the median values of the experimentally obtained distance-to-FA-edge distributions per cell (over multiple FAs) and those corresponding to the simulated data. We then defined the edge-proximity-factor (EPF) as the relative difference between experimental and simulated distributions, where negative EPF values correspond to preferential nanocluster enrichment at FA edges, positive EPF values to nanocluster depletion at FA edges, and EPF = 0 corresponds to random location of nanoclusters with respect to the FA edges (**Fig. 3B**). Moreover, to statistically determine whether the distribution of EPF values was different from zero (the EPF value expected from a random distance-to-the-FA-edge distribution) we ran a paired Student's t-test for each protein comparing the experimental with the corresponding simulated distributions. This analysis showed that most proteins exhibit some degree of enrichment at the edges of the FAs (i.e., negative EPF values), although their magnitude was markedly different amongst them (**Fig. 3B**). The highest enrichments at the FA edges were found for the full population of $\alpha_5\beta_1$ nanoclusters and the active β_1 sub-population (~30% closer to the FA edge than if uniformly distributed) and for talin, whereas the enrichment at the FA edges of the total $\alpha_v\beta_3$ population and paxillin was much more modest. Notably, the active β_3 sub-population and vinculin showed no preferential enrichment at the FA edges (**Fig. 3B**).

To establish the characteristic distances at which nanocluster enrichment occurs with respect to the FA edges, we next computed the difference between the experimental and simulated distributions, i.e., the variation from random, as a function of the distance to the FA edges, for each of the examined proteins (**Fig. 3C, D**). Here, positive values correspond to local nanocluster enrichment at a given distance to the FA edge, whereas negative values indicate nanocluster

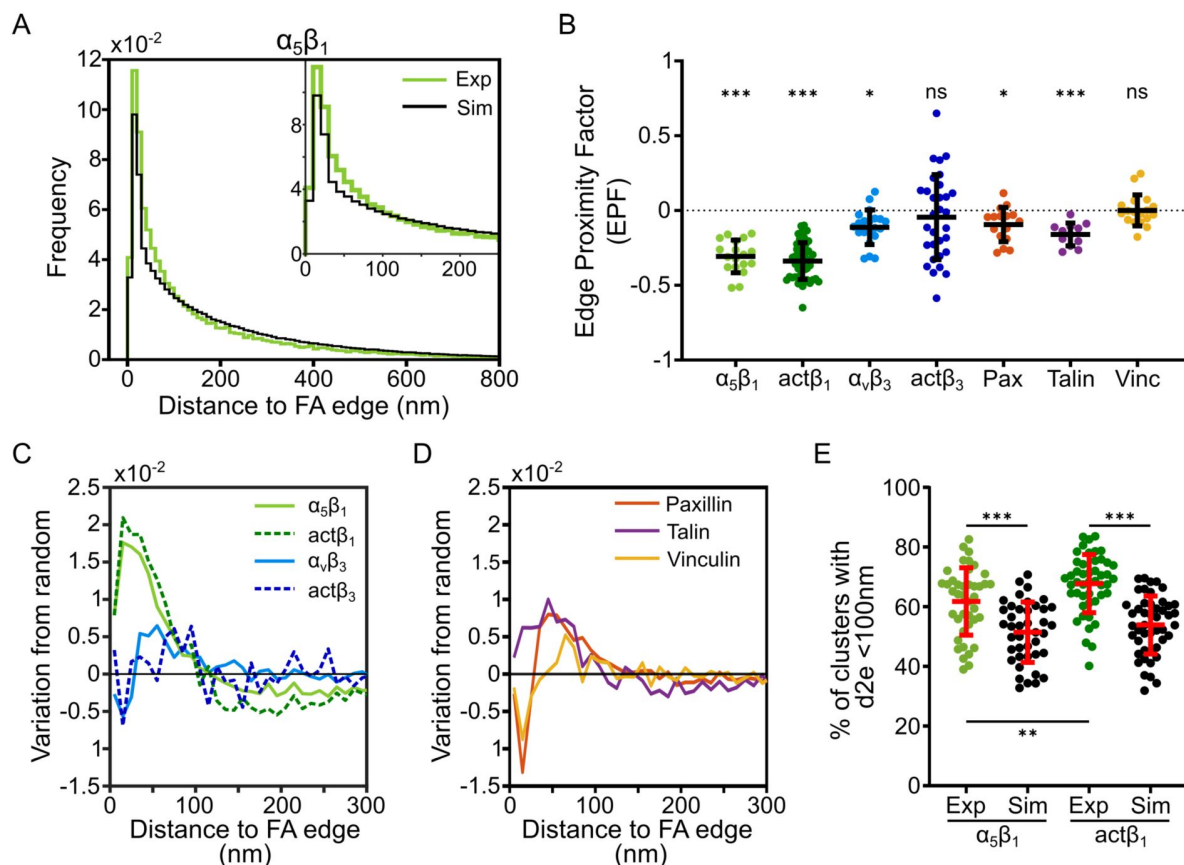


Figure 3

Integrin $\alpha_5\beta_1$ nanoclusters are preferentially distributed on the edges of FAs.

(A) Distribution of distances from the center of mass (CoM) position of each individual $\alpha_5\beta_1$ nanocluster to its closest FA edge obtained from DBSCAN-analyzed STORM images. The histogram represents the frequency distribution over all cells. Green lines correspond to the experimental data (Exp) and the simulated uniform distribution (Sim) is shown as a black histogram. Bin width is 10 nm. The inset shows the distribution for the first 250 nm. (B) Edge Proximity Factor (EPF) per cell for each of the proteins investigated. Data are shown as median values per individual cell (color dots). The mean $\pm$ st. dev. over all the cells are superimposed as black lines. Statistical significance between experimental and simulated data was tested using a paired t-test, where ns = not significant, $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. (C, D) Variation of the nanocluster distances from the FA edges with respect to random organization, as function of the distance to the FA edges for $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrin nanoclusters (C) and for the adaptor proteins (D). (E) Percentage of nanoclusters proximal to FA edges ($< 100\text{nm}$) for the total and active β_1 nanocluster population (color dots) together with results from simulations of random organization (black dots). Each dot corresponds to the median value per cell, each cell containing multiple FAs. The mean $\pm$ st. dev. over all the cells is superimposed as red lines. Statistical significance between experimental and simulated data was tested using a paired t-test, and to test the significance between $\alpha_5\beta_1$ and $\alpha_v\beta_1$ the unpaired t-test was used, assuming a Gaussian distribution of data, where ns = not significant, $p > 0.05$; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

depletion. Our results show a clear difference between both integrins, with $\alpha_5\beta_1$ nanoclusters being enriched close to the edge of FAs (**Fig. 3C**, continuous green line) whereas $\alpha_v\beta_3$ nanoclusters showed a rather uniform distribution, regardless of their distance to the FA edges (**Fig. 3C**, continuous blue line). Moreover, the preferential enrichment of β_1 but not of β_3 nanoclusters at the FA edges was retained when specifically analyzing the active integrin conformations (**Fig. 3C**, dashed green and blue lines, respectively).

We next investigated how adaptor protein nanoclusters distribute with respect to the FA edge using a similar approach. We found that both talin and paxillin nanoclusters showed enrichment in a region between 50–150 nm from the FA edges, although somewhat less pronounced as compared to $\alpha_5\beta_1$ nanoclusters (**Fig. 3D**), whereas vinculin nanoclusters showed no preference to reside proximal to the FA edges. Finally, we extracted the percentage of the total $\alpha_5\beta_1$ and active β_1 nanoclusters that reside within distances shorter than 100 nm from the FA edges and compared the results to those from simulations of random organization (**Fig. 3E**). Notably, a significantly larger percentage of active β_1 nanoclusters as compared to the total $\alpha_5\beta_1$ nanocluster population and to the simulations, locate at the periphery of FAs (**Fig. 3E**).

Together, these results strongly indicate that integrin nanoclusters spatially organize in a rather distinct fashion within FAs, with $\alpha_5\beta_1$ nanoclusters localizing preferentially at the edges of the FAs and $\alpha_v\beta_3$ nanoclusters distributing randomly inside FAs. Moreover, the preferential enrichment of active β_1 nanoclusters together with talin at the FA edges suggests that β_1 activation is spatially regulated inside FAs.

Active β_1 nanoclusters form multiprotein hubs at FA edges

Since activation of integrins involves engagement with their adaptors paxillin, talin and vinculin, we sought to investigate whether the active β_1 nanoclusters preferentially located at the FAs periphery also reside in proximity to their adaptor proteins. For this we first obtained the inter-nanocluster NND (iNND) distributions between active β_1 nanoclusters and each of the adaptor nanoclusters (**Fig. S4B**). We calculated the median of each iNND distribution and used this value as a threshold to classify pairs of active β_1 -adaptor nanoclusters as being close (i.e., iNND < median) or far (iNND > median) from each other. Next, we separately computed the distance-to-the-FA edge for the active β_1 nanoclusters that were found close or far from their partner and compared the results to simulated data (**Fig. 4A** and *inset*). Having these plots, we then calculated the variation from random (i.e., difference between experimental data and simulations of random organization) as a function of the distance to the FA edge, for both active β_1 nanoclusters close to their partner and active β_1 nanoclusters far from their partner (**Fig. 4B** and **Fig. S4C**). This analysis shows that active β_1 nanoclusters preferentially locate the FA edges regardless of their proximity to talin nanoclusters (**Fig. 4B**) and to other adaptors (**Fig. S4C**). In contrast, the same analysis on active β_3 nanoclusters shows a random distribution inside of FAs irrespective of their proximity to talin or other adaptor nanoclusters (**Fig. 4C**).

Finally, we enquired whether adaptor protein nanoclusters (paxillin, talin, and vinculin) close to active β_1 nanoclusters also locate closer to the FA edges. Notably, we found that all of these adaptor nanoclusters close to active β_1 nanoclusters are also enriched on the FA edge region (~150–200 nm from the edge), whereas adaptor nanoclusters far from active β_1 appear to be more uniformly distributed (**Fig. 4D–F**). Taken together, our data reveal a differential mesoscale organization between active β_1 and β_3 nanoclusters inside FAs, with active β_1 nanoclusters being enriched at the periphery of FAs alongside adhesion adaptor nanoclusters, while active β_3 nanoclusters remain uniformly distributed throughout FAs, lacking peripheral enrichment.

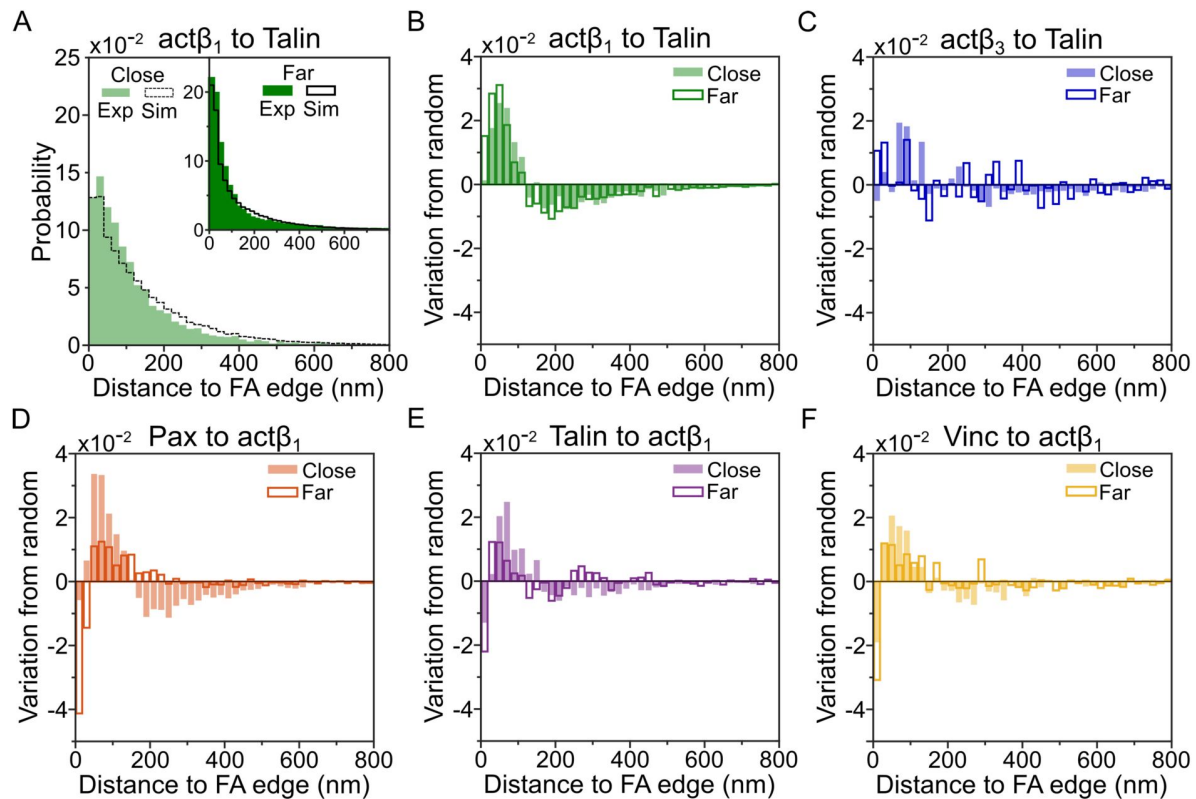


Figure 4

Active β_1 nanoclusters form multiprotein hubs at the FA edges.

(A) Population of active β_1 -talin nanocluster pairs as a function of their distance-to-the-FA edge. The data plotted in light green correspond to active β_1 nanoclusters located close to talin (i.e., iNND < median), while the inset dark green corresponds to active β_1 nanoclusters located far from talin (i.e., iNND > median). iNND distributions are shown in Fig. S4B. Black lines are for simulated data sets. Bin width is 20 nm. (B) Variation from random for active β_1 -talin nanocluster pairs as function of their distance-to-the-FA-edge. Filled bars correspond to active β_1 nanoclusters located close to talin (iNND < median) and empty bars correspond to active β_1 nanoclusters located far from talin (iNND > median). (C) Similar to (B) but for active β_3 nanoclusters. (D-F) Variation from random for paxillin (D), talin (E), vinculin (F) active β_1 nanocluster pairs, as function of their distance-to-the-FA-edge, with data segregated in two different distributions according to the pairs proximity. Bin width is 20 nm.

Discussion

Here, we have combined dual-color super-resolution optical microscopy together with a robust set of analytical tools to provide insights on the lateral spatial organization of different adhesion proteins inside FAs. Earlier reports showed that nascent adhesions (NAs) are formed due to integrin nanoclusters that either rapidly disassemble at the plasma membrane without force or further mature into FAs in response to mechanical forces (35, 48). Our results now show that even in the presence of mechanical forces (i.e., inside FAs), integrin nanoclusters remain segregated from each other while maintaining their sizes and number of molecules. Similar integrin nanocluster properties were also found outside adhesions, reinforcing the notion that integrin nanoclustering is force-independent (35). Interestingly, we found that the main cytoplasmic adaptors, paxillin, talin and vinculin, also form nanoclusters of similar size and number of single molecule localizations as the integrins, regardless of whether they locate inside or outside FAs. The presence of adhesion protein nanoclusters outside of FAs is intriguing. However, immobilized (active) integrins have been previously observed outside of FAs (25, 49). We suggest that these isolated nanoclusters may represent transient nascent adhesions forming on the cell membrane. Although STORM images do not allow to directly extract the actual number of proteins contained in nanoclusters given the primary and secondary antibody labeling strategy used, the similarity in terms of number of single molecule localizations amongst integrins and their adaptor partners suggest that these nanoclusters might be stoichiometrically similar. Together, our results strongly indicate that adhesion proteins organize as universal nano-hubs, functioning as modular adhesion units to serve multiple integrin-dependent functions related to adhesion and migration but also to mechanosensing and mechanotransduction.

Consistent with previous results (36), our data revealed that active and inactive $\alpha_5\beta_1$ nanoclusters coexist inside FAs but are fully segregated from each other. Remarkably, $\alpha_v\beta_3$ also exhibits similar nanoscale segregation, with total population-to-active integrin ratios indicating a similar segregation between active and inactive β_3 . Furthermore, earlier data from our group demonstrated that the integrin $\alpha_1\beta_2$ expressed on leukocytes also segregates into active and inactive nanoclusters (50, 51). Thus, lateral segregation between active and inactive integrin nanoclusters may represent a common feature to spatially coordinate integrin activation. How could such cooperativity in integrin nanocluster activation arise? Our data show a nearly 1:1 stoichiometry in the number of active integrin nanoclusters, and the main cytoplasmic adaptors talin and vinculin, known to activate integrins by their engagement to the actin cytoskeleton. The availability of a similar number of adaptor nanoclusters containing comparable number of molecules as the integrin nanoclusters suggests that inactive integrin nanoclusters are poised for their collective nanocluster activation, which is achieved upon the concurrent engagement of talin and vinculin nanoclusters (and most probably also paxillin). This mechanism has been shown to operate in NAs in which concurrent recruitment of talin and vinculin prior to force generation is required for integrin activation and further maturation of NAs (52). In addition, our data reveals that there is a large percentage of inactive $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins inside FAs and very little spatial overlap between integrins and main adaptors. As integrin activation is highly dynamic and transient, we propose that this pool of inactive integrin nanoclusters is required to increase the probability of new engagements between integrins and their adaptors once the active integrin nanoclusters become disengaged, stabilizing as a whole the FAs for lifetimes much longer than the binding-unbinding dwell-times of integrins with their partners.

Quantification of super-resolution data showed a higher number of $\alpha_5\beta_1$ nanoclusters compared to their $\alpha_v\beta_3$ counterparts inside FAs. The relative percentage of active β_1 nanoclusters is also higher than that of $\alpha_v\beta_3$ and moreover these two integrin nanoclusters never intermix inside FAs. Importantly, active β_1 nanoclusters preferentially locate at the periphery of FAs and in proximity with paxillin, talin and vinculin nanoclusters, while active β_3 are more uniformly distributed

inside FAs. The differential degree of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ in terms of expression, activation state, and lateral nano- and meso-scale segregation, strongly indicates that integrin activation and their distinct functions inside FAs are spatially regulated (**Fig. 5**). Based on our results we postulate that (i) $\alpha_5\beta_1$, through the ring on the FA periphery and its close proximity to paxillin, talin and vinculin nanoclusters, primarily functions for substrate attachment in its high tensional state, engaging with actomyosin filaments; (ii) $\alpha_v\beta_3$ and $\alpha_5\beta_1$ integrin nanoclusters in the central region of the FA undertake the role of mechanotransduction, being more dynamic in terms of activation and inactivation states and most probably engaged with shorter branched actin filaments; and (iii) the difference in these roles is controlled by the distinct mechanical forces exerted on the integrins, depending on their location inside FAs, and for $\alpha_5\beta_1$ perhaps also on its different tensional states (53).

Intriguingly, our analysis showed that active β_1 nanoclusters preferentially locate the FA edges regardless of their proximity to talin nanoclusters (**Fig. 4B**). These data are striking since active integrins are expected to be bound to both its ligand and talin, and therefore had anticipated the whole active β_1 population to be in close proximity to talin nanoclusters. We can envisage two plausible explanations for active β_1 to be far from talin. One possible scenario is the existence of other adaptor protein(s), aside from talin, able to bind to active β_1 . Such a candidate might be tensin3, which has been recently shown to compete with talin to bind $\alpha_5\beta_1$, during the formation of FAs from FAs (54). A second possibility to explain our observations is the nature of talin itself. As a long rod-like protein, talin stretching experiments have shown that the N- and C-termini can be separated up to 250 nm due to actomyosin activity (55). Since we detect talin by labeling it in its C-terminal domain, which is opposite to the integrin binding side, it is possible that active β_1 is engaged to extended talin even for inter-cluster distances larger than the median iNND. Further experiments using specific antibodies against tensin3 as well as labeling talin at the closest proximity to the integrin binding side would clarify the adaptor partners associated to active β_1 nanoclusters at the periphery of FAs.

In summary, in contrast to the classical textbook view where FAs are pictured as dense and homogenous assemblies of integrins fully engaged to their cytoplasmic adaptors, our work reveals a highly complex organization inside FAs, where regulation of different integrins and their activation state are highly coordinated in space. We believe that such spatial regulation provides the means for different integrins to synergize and/or diversify their functions in terms of adhesion, migration, mechanosensing and mechanotransduction. Earlier reports showed that proteins such as KANK1, KANK2, and β_1 integrins are enriched at the edges of adhesions, with KANK2 reducing the force exerted by the cells through its binding to talin (56, 57). KANK proteins have also been implicated in enabling communication between microtubules and actin by binding to talin and liprin- β_1 at FA edges (58). More recent data suggest that the localization of KANK1 around FAs is driven by liquid-liquid phase separation (59). Our findings together with those mentioned above suggest that the $\alpha_5\beta_1$ nanocluster belt is an important feature of FAs and may be critical for their function. However, further research is needed to fully understand how the organization of these key proteins contributes to their roles FAs. p130Cas- and FAK-dependent phase separation has been also suggested to promote integrin clustering and NA assembly (60). Whether phase separation also contributes to the preferential organization and activation of β_1 at the FA edges as we reveal here is an intriguing topic that deserves further investigation.

Materials and methods

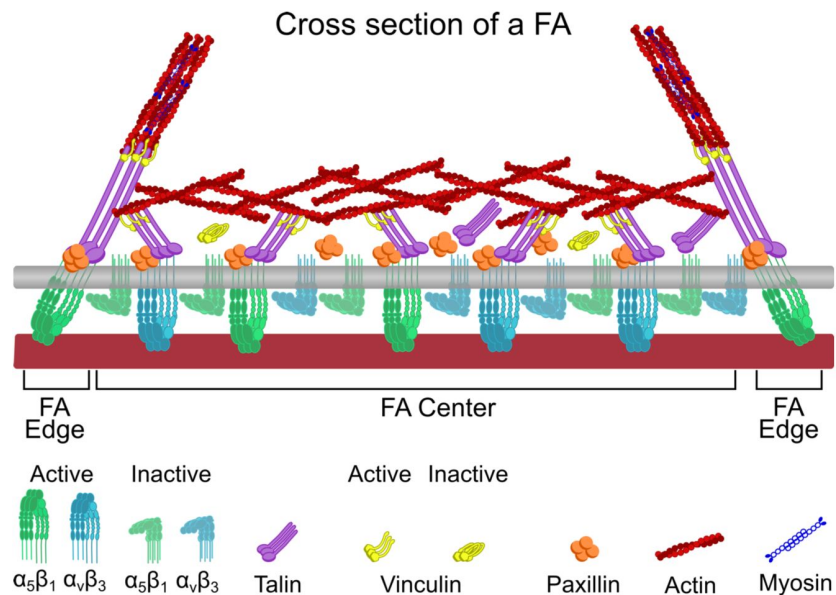


Figure 5

Spatial regulation and activation of integrin nanoclusters and protein adaptors inside focal adhesions.

$\alpha_5\beta_1$, $\alpha_v\beta_3$, and main protein adaptors organize as small nanoclusters inside FAs physically segregated from each other. A considerable high fraction of $\alpha_5\beta_1$ and $\alpha_v\beta_3$ integrins remain in an inactive state and do not intermix at the nanoscale. Active β_1 nanoclusters (green) on the FA edge reside in close proximity to the adaptor protein nanoclusters and thus possibly engaged to actomyosin filaments (red and blue). Integrin nanoclusters in the FA center are more randomly distributed with respect to one another, most probably engaged with branched cortical actin. According to our results we propose that $\alpha_5\beta_1$ nanoclusters located at the FA periphery could be primarily involved in cell substrate attachment, whereas $\alpha_5\beta_1$ and $\alpha_v\beta_3$ nanoclusters at the center of FAs engage in mechanosensing and mechanotransduction. For simplification, we show a cross-section of a FA where active β_1 integrins at the edge are engaged with talin. However, our data also show the presence of active β_1 nanoclusters spatially separated from talin and thus possibly engaged to other binding partners linking β_1 to the actin cytoskeleton (not shown in the cartoon for the sake of clarity).

Sample preparation and immunolabeling

HFF-1 cells (human foreskin fibroblasts) were obtained from ATCC and used with low passage number (passage <15). Cells were maintained in DMEM supplemented with 15% FBS and cultured in a humidified atmosphere at 37° C and 5% CO₂. Experiments were performed in cells seeded on 8-well plates (#1 Lab-Tek, Thermofisher, #155411) that were previously coated with fibronectin (FN) at full substrate coverage. FN coating was performed by incubation with 10 µg/ml FN for 30 min at room temperature (RT). We note that FBS can contain vitronectin (VN) and other proteins that may act as ligands of integrins. However, since the glass slides are saturated with FN prior to incubation with cells and media, FN remains the dominant ligand in our experiments. After seeding, cells were allowed to spread for 24 h at 37° C and 5% CO₂, after which they were fixed by using 4% PFA in PBS for 15 min at RT. Cells were permeabilized using 0.1% Triton X-100 in PBS for 10 min and blocked with 3% BSA in PBS for 40 min. Primary antibodies against integrins $\alpha_5\beta_1$ and $\alpha_v\beta_3$, total population, or specific for the active conformation of integrins β_1 and β_3 , and total populations of adaptor proteins, paxillin, talin (actin binding C-terminal epitope), or vinculin, were diluted in blocking buffer and added to the cells for 30 min at RT. After washing with PBS, secondary antibodies that were conjugated in-house with dye pairs Alexa Fluor 405-Alexa Fluor 647 (405-647 channel) or Cy3B-Alexa Fluor 647 (Cy3-647 channel), were added to the cells for 30 min at RT (see Tables S1-S3 for details on reagents and antibodies used). After washing with PBS, cells were stored in PBS until imaged.

Image acquisition

STORM data were obtained on a commercial Nikon Eclipse Ti system (N-STORM), equipped with a 100x oil objective with NA 1.49 using TIRF illumination. The detector was an ANDOR technology EMCCD iXon 897 camera, with a 256×256 pixel ROI and a pixel size of 160 nm. The system has an Agilent technologies laser box with wavelengths of 405 nm, 560 nm and 647 nm. STORM was performed by incubating cells in an imaging buffer containing 100 mM of the reducing agent cysteamine mercaptoethylamine (MEA) (77 mg of MEA in 1 ml of 360 mM HCl, stock solution 1 M), the enzymatic oxygen scavenger Glox solution (0.5 mg/ml glucose oxidase, 40 µg/ml catalase), and 6.25% glucose in PBS. Details of the STORM imaging buffer reagents are found in Table S4. Dual color imaging was carried out cyclically and sequentially. In brief, we first detected the 405-647 channel in a 5-frame cycle consisting of 1 activator frame (405 nm laser on) followed by 4 reporter frames (647 nm laser on), followed by the Cy3-647 channel detection sequence, similarly composed of 1 activator frame (560 nm laser on) followed by 4 reporter frames (647 nm laser on). The frame rate was 50 Hz (20 ms per frame) for all frames. Imaging was carried out by cyclically repeating this 10-frame sequence until all the reporter dyes were exhausted (typically ~70,000 frames).

Single molecule detection, localization

precision, and STORM image reconstruction

Dual-color STORM image reconstruction was done using Insight3 software, provided by Bo Huang (UCSF, initially developed in Xiaowei Zhuang's lab, Harvard University). In each imaging frame, individual molecules were identified using intensity thresholding (gray levels set between 600-60,000), then each intensity profile was fitted with a 2D Gaussian function to identify the central position, i.e., the coordinates that represent the spatial localization of that fluorescent molecule. The analysis was run over the entirety of each data set, resulting in a final output that is a list of x,y coordinates (localizations) corresponding to each individual molecule that has been identified. Insight3 software allows for drift correction using the correlated function of the image, for details see (61 [link](#)). In this multi-color imaging scheme, the main source of crosstalk is the nonspecific activation of the Alexa 647 by the 647 nm laser (62 [link](#)), which is activator-independent and can occur in any frame. We implemented statistical crosstalk correction where the Insight3 software analyses the full image in 50×50 nm regions of interest (ROIs) and determines if each ROI is predominantly one color or the other. Based on the acquisition design (in our case, one activator

frame followed by four reporter frames) the software determines the probability of detecting a nonspecific reporter molecule (62). To minimize color cross-talk, only those localizations detected in the second frame of each five-frame STORM imaging cycle (that is, the first of the four reporter frames) were considered for future analytical steps. Since each single molecule can experience numerous blinking events, the variation of the x,y coordinates from frame to frame determines the effective resolution (63). To measure this, we detected multiple localization events arising from individual molecules deposited on glass and calculated the standard deviation of the localization positions. We measured the effective resolution of our system to be ~15 nm. These localizations detected from nonspecific antibodies on glass were also used to set a threshold for what we considered to be a nanocluster of proteins. We found that the antibodies on glass had a median number of localizations of 5 (for channel 1) and 3 (for channel 2). With the aim of discarding all localizations coming from nonspecific binding or binding to individual proteins we set the threshold at 10 localizations per nanocluster.

Generation of masks and classification of the localizations belonging to FAs, fAs or outside adhesions

A Fiji plugin, developed by the Lakadamyali lab, was implemented as an initial step to split the localization data from the STORM images into separate data sets corresponding to the localizations contained in certain ROIs (64). First, the data from the two channels of the reconstructed STORM image were imported and the localizations rendered together in a 1024×1024 pixels image (Fig. S1A). Next, we applied an intensity threshold, until the adhesion areas were visually detected, a Gaussian blur of the image (by 2 pixels), and a second threshold filter until a smooth binary mask was obtained (Fig. S1B). The final mask retaining only the adhesion areas was obtained by manually selecting the regions corresponding to the adhesion complexes (identified by visual inspection based on their size and location). In the first steps, this Fiji plugin creates separate files containing the localizations that fall within the selected ROIs: all adhesions, glass, and outside adhesions (i.e., cell membrane that contain the remaining areas of the image that do not correspond to all adhesions or glass). In a second step the “all adhesions” file was again analyzed by the same plugin and all adhesions were split into FAs and fAs and rendered with Insight3. Fig. S1C-F shows the resulting rendered localizations for the different regions. These final two-color files for different cell regions were then individually and independently analyzed.

Image analysis

Clustering detection using the DBSCAN algorithm

DBSCAN is a density-based data clustering algorithm designed to be executed on arbitrary data sets with minimal input from the user (40). The algorithm works by identifying arbitrary-shaped clusters of data points using two user-defined input values: epsilon (ϵ), which is a radial distance that serves to define a local neighborhood from a randomly selected initial data point; and $N_{\min}$, which is the minimum number of points required to be within the initial point's neighborhood, so it is considered part of the cluster seed. For our image analysis, spots on glass outside the cell regions were analyzed using $\epsilon = 20$ nm and $N_{\min} = 2$, in order to define a minimum threshold for the number of localizations per nanocluster. Important to note that we carried out this analysis on glass areas, next to the cells, found in all of our STORM images. In this way we ensure having the same imaging conditions for in-cell nanoclusters and for spots-on-glass, however, at the expense of having a broader distribution of localization per spot-on-glass due to the presence of cell debris on the glass. When analyzing the nanoclusters found in ROIs within the cell, the DBSCAN parameters were set to $\epsilon = 20$ nm and $N_{\min} = 3$ localizations, which are common values used in the field (36, 41). Nevertheless, since multiple blinks of a fluorophore might result in artifactual clusters, we only considered true nanoclusters as those containing 10 or more

localizations. Examples on the identification of different nanoclusters by DBSCAN are shown in Fig. S1G. The algorithm also provides the center of mass (CoM) coordinates of each nanocluster together with the number of localizations per nanocluster and their diameter.

Nearest neighbor distance (NND) analysis.

To establish how the protein nanoclusters are distributed relative to one another, we used the MATLAB (MathWorks) function *knnsearch*. This function computes for each nanocluster the NND, which is the shortest distance from the CoM of a given nanocluster to its closest CoM nanocluster neighbor. We also used the same function to determine the *i*NND (i.e., NND between nanoclusters of different proteins). To measure the percentage of nanoclusters that overlap (colocalization fraction), we first computed the edge-to-edge distance between nearest neighbors (EED-NN) by subtracting from the NND the radii of the two-corresponding near-neighbor nanoclusters. From the EED-NN distributions we calculated the percentage of occurrences with EED-NN values below zero. The *knnsearch* function was also used to extract the distance-to-the-FA edge, i.e., the shortest distance between nanoclusters and the edge of adhesions. To that end, we computed the NND between the CoM of each nanocluster to the center coordinates of the list of pixels that fall on the edge of the FA mask.

Computational generation of uniform nanocluster distributions

To assess whether the distance distributions obtained from the experimental data (NND, EED-NN and distance-to-the-FA-edge), correspond to a preferential type of organization or to a uniform distribution, we performed simulations by generating *in silico* histograms of non-overlapping and randomly distributed nanoclusters inside FAs. For this, we first extracted the exact number and size of nanoclusters per adhesion from the experimental data. Although our initial approach was to computationally generate the same number of nanoclusters and distribute them uniformly over the adhesion masks regardless of their physical size, we found that this approach was not computationally suitable. The reason for it is that in many cases the large nanoclusters did not fit (without overlapping) in the remaining space available within the adhesions after randomly positioning the large majority of smaller nanoclusters. We thus opted for classifying the nanoclusters according to their physical size and then start our procedure by first positioning the largest nanoclusters and later all the remaining smaller ones. In addition, as our experimental data renders well segregated nanoclusters, we imposed in our algorithm the condition that nanoclusters should not overlap with one another. Fig. S5 shows the different steps followed in order to generate distributions of random nanoclusters inside FAs.

Statistical analysis

Statistical analysis was carried out using GraphPad Prism (version 9.12). The details of the individual statistical tests as well as the level of significance along with p-values are explicitly provided in the figure legends. No predetermined statistical method was employed for establishing the sample size, as the sample sizes were not selected based on a predefined effect size. In this context, multiple experiments were independently performed, employing various sample replicates, as detailed in Table S5.

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request. The codes used in this study are available from the corresponding author upon reasonable request.

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Additional information

Author Contributions

S.K prepared the biological samples, performed STORM imaging and analyzed the data. N.M. developed algorithms for data analysis and contributed to the analysis. F.C. and M.F.G.-P. supervised the research. S.K., F.C. and M.F.G.-P conceived and discussed the experiments, interpreted the data, and wrote the paper.

Additional files

Supplementary Material [↗](#)

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Reviewer #1 (Public review):

Summary:

In recent years, it has become increasingly evident how beautifully intricate IAC are at the nanoscale. Studies like the one presented here that shed light on the precise inner organisation of IAC are thus quite important and relevant in order to obtain a better in-depth understanding of IAC functioning and the contribution of different integrin subtypes to cell adhesive and mechanotransductive processes.

Interestingly, the authors found a distinct localisation of $\alpha 5 \beta 1$ and $\alpha V \beta 3$ integrin nanoclusters within focal adhesion of human fibroblasts, with $\alpha 5 \beta 1$ integrin nanoclusters being at the periphery of IAC and $\alpha V \beta 3$ integrin nanoclusters randomly distributed. Furthermore, a surprisingly high percentage of inactive integrins within IAC and relatively low spatial integrin colocalisation with adaptor proteins has been shown.

Strengths:

This is a very thoroughly performed STORM-based assessment of the nanodistribution of $\alpha 5 \beta 1$ and $\alpha V \beta 3$ nanoclusters within IAC (and outside). The image quality is outstanding, and the authors have meticulously executed the experiments and the image analyses.

Weaknesses:

The only weakness is maybe that the manuscript remains descriptive. However, the high quality of the "description" of the nano-organisation of IAC by this scrupulous study is really important to better understand the inner workings of IAC. It provides a very solid foundation to look deeper into the (patho)physiological implications of this organisation, see recommendations (which are rather suggestions in this case).

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Reviewer #2 (Public review):

Summary:

In this study, dual-color super-resolution microscopy analysis was performed to study the co-operation between integrins and focal adhesion proteins in human fibroblast cells. The study focused on two integrins which have been previously found to be mainly responsible for focal adhesions, namely $\alpha 5 \beta 1$ and $\alpha v \beta 3$.

Specifically, the study tried to shed light on the nanoclustering of integrins in focal adhesions.

In the current study, more integrin nanoclusters were observed in focal adhesions compared to other cell-matrix adhesion structures. The study revealed that both $\alpha 5 \beta 1$ and $\alpha v \beta 3$ form nanoclusters, and those appear segregated from each other. While $\alpha v \beta 3$ nanoclusters organize randomly inside focal adhesions regardless of their activation state, $\alpha 5 \beta 1$ nanoclusters, and particularly the nanoclusters containing $\beta 1$ -integrin in active conformation, preferentially organized at the edges of focal adhesions. The nanoclusters formed by each integrin were similar in size.

Cytoplasmic adapter proteins appeared less in nanocluster assemblies, suggesting that integrin nanoclusters are also forming without the studied cytoplasmic adapter proteins (talin, vinculin, paxillin). Active integrins were identified with the help of conformation-specific antibodies, and this enabled us to study the colocalization between integrins and their cytoplasmic adapter proteins. This analysis revealed that activated integrins are strongly engaged with adapter proteins

Strengths:

The study stems from the thorough computational modelling of the nanoclusters, which enables quantification of the behavior of the clusters, including their mesoscale distribution.

The study strengthens the view that $\alpha 5 \beta 1$ and $\alpha v \beta 3$ have specific functions in focal adhesions, $\alpha 5 \beta 1$ nanoclusters localizing preferentially on focal adhesion edges. The study also revealed that nanoclusters localized at the edges of focal adhesion were enriched for talin and paxillin but not for vinculin.

Analysis of adaptor protein nanoclusters (paxillin, talin, and vinculin) revealed that all adapter protein nanoclusters studied here close to active $\beta 1$ nanoclusters are enriched on the focal adhesion edge region, whereas integrin adaptor nanoclusters far from active $\beta 1$ appear to be more uniformly distributed.

Importantly, the current study suggests that integrin subtype-specific nanoclusters are not only present at an early stage of adhesion formation, but integrin nanoclusters remain segregated from each other also in mature focal adhesions, maintaining their sizes and number of molecules.

Interestingly, the study revealed that selected cytoplasmic adaptors (paxillin, talin, and vinculin), also form nanoclusters of similar size and number of single molecule localizations

as the integrins, regardless of whether they locate inside or outside focal adhesions. The adapter nanoclusters are enriched in the focal adhesion "belt", colocalizing with the active $\alpha 5 \beta 1$ integrin nanoclusters.

Weaknesses:

The current study is highly dependent on the antibodies. It is possible that antibodies containing two binding sites for antigen influence the nanoscale organization (and also activation) of the receptors. Control experiments to study the possible contribution of antibodies to the measured outcome should be performed to verify the main findings. One possible approach could be to use fluorescently tagged integrins available. Alternatively, integrins (or adapter proteins) could be tagged with a small ligand and detected using a monovalent binder.

Only a limited number of integrin adapter proteins were investigated. Given the high number of identified adapter proteins, this is an understandable choice. However, it would be fascinating to understand if the nanoclusters of inactive integrins are dominantly bound with a certain adapter protein, such as tensin.

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Reviewer #3 (Public review):

Summary:

In their study, the authors reveal using dual-color super-resolution STORM microscopy modality and immunolabeling in fixed adherent cells, that $\beta 1$ and $\beta 3$ integrins as well as adaptors (paxillin, talin and vinculin) are all organized in nanoclusters of similar size (50nm) and molecular density (20 copy number) inside FAs but also outside. Using activity-specific immunolabeling of $\beta 1$ and $\beta 3$ integrins, they revealed that active integrin subpopulations were both clustered but in distinct exclusive nano-aggregates in agreement with Spiess et al. (2018). Once more, the "active" integrin nanoclusters displayed similar properties in terms of size and molecular density, suggesting that molecular organization in nanoclusters is an intrinsic property of integrins in plasma membrane multimerizing independently of their location (inside or outside FAs), their level of activation, or their connection to the cytoskeleton. Then the authors followed up by analyzing at the mesoscale how these "universal" nanoclustered adhesive units are distributed spatially. Inspecting the surface density of nanoclusters revealed that the density of integrin nanoclusters in FAs was 5x larger, compared to integrin nanoclusters outside adhesions. Interestingly, whereas the density of total integrin nanoclusters was 2-4x larger than adaptor nanoclusters, the density of "active" integrin nanoclusters stoichiometrically matches that of talin and vinculin nanoclusters, and was slightly outnumbered by paxillin nanoclusters. These findings suggest that inside FAs, among the total number of integrin nanoclusters, the subset of "active" integrin nanoclusters could be engaged with "adaptor" nanoclusters on a 1:1 ratio. Using analysis of the nearest neighbor distance (NND) between distinct integrin clusters and each of the adaptors, the authors report that they found negligible spatial colocalization of integrins with these adaptor proteins and that spatial segregation is essentially determined by the density of nanoclusters within the FAs. As authors reported that $\alpha 5 \beta 1$ and $\alpha v \beta 3$ do not intermix at the nanoscale, the authors finally highlighted how $\alpha 5 \beta 1$ and $\alpha v \beta 3$ distinct nanoclusters are differently organized and segregated inside FAs. Adapting the NND analysis in order to inspect how far the nanoclusters are from the edges of FAs they are located in, authors revealed that $\alpha 5 \beta 1$ but not $\alpha v \beta 3$ integrin nanoclusters are enriched on FA edges and that similar FA edge-enriched distribution for "active" $\alpha 5 \beta 1$ and adaptor protein nanoclusters was found for talin and paxillin but not vinculin. The latter results suggest that FA edges could constitute multiprotein hubs for enhanced colocalization and activation for $\alpha 5 \beta 1$

integrin nanoclusters and adaptors such as talin and paxillin. Unfortunately NND analysis could not confirm this enhanced colocalization hypothesis.

General Assessment:

While the study presents some valuable findings, it reads currently as a compilation of intriguing but preliminary observations derived primarily from a single methodology (dual-color STORM and DBSCAN clustering analysis). As the initial findings often lack confirmation through additional data analysis (such as the NND analysis the authors used), there's a critical necessity to bolster the methodological approach. This should involve replicating the main findings using alternative single-molecule super-resolution techniques (such as quantitative DNA-PAINT) or employing different clustering analytical tools (such as voronoi-tessellation). Furthermore, the manuscript feels incomplete, focusing solely on describing molecular organization without offering substantial insights into how these observations correlate with the regulation, activation, and functionality of integrins at the cellular level.

The manuscript presents extensive datasets and utilizes methodologies in which the investigators demonstrate expertise. Nevertheless, there's uncertainty regarding the novelty and broad appeal of the findings. For instance, the observation of integrin nanoclustering has been previously reported in several publications (e.g., Changede et al., *Dev Cell* 2015; Spiess et al., *JCB* 2018; Fujiwara et al., *JCB* 2023). Similarly, the accumulation of specific proteins at the periphery of FAs has been documented elsewhere (e.g., Sun et al., *NCB* 2016; Stubb et al., *NatComm* 2019; Nunes-Vicente TCB 2023), as well as the differential dynamic organization of $\alpha 5 \beta 1$ and $\alpha v \beta 3$ integrins inside FAs (e.g., Rossier et al., *NCB* 2012). Beyond the universal organization of adhesive proteins, there's a need to identify novel insights that significantly advance the field. One potential avenue could involve pinpointing the molecular determinant controlling the FA edge enrichment of active $\alpha 5 \beta 1$ integrins and talin nanoclusters. For instance, could there be an interplay between $\alpha 5 \beta 1$ and $\alpha v \beta 3$ integrin nanoclusters visible on one's organisation when suppressing the other using deletion (KO) or depletion (SiRNA)? Also, could KANK, which also exhibits enrichment and regulates talin activity (e.g., Sun et al., *NCB* 2016), play a role in this process? Identifying the molecular players that regulate even partially the mesoscale organization of nanoclusters of proteins would really benefit the breadth of this manuscript.

Echoing the previous concern, the manuscript described a novel and rather surprising finding related to molecular clustering of adhesion proteins. Indeed, the fact that nanoclusters exhibit uniform size and molecular density regardless of the protein type, location, or activation level is indeed surprising and raises many questions about the methodology used to assess molecular clustering. I feel that the description and characterization of integrin nanoclusters appear incomplete and need to be expanded by comparing different analytical strategies for protein clustering. Furthermore, a lack of the manuscript in its actual form concerns the quantification of integrin numbers inside the observed nanoclusters. I agree that the path from optical microscopy to protein stoichiometry quantification is hard and full of drawbacks. But the authors do not fully address these issues that are extremely important when discussing protein nanoclustering. This quantitative aspect should be discussed.

First, it is crucial for the authors to carefully examine and discuss in their manuscript whether there are any potential biases or limitations in the experimental techniques (dual-color STORM) or data analysis methods employed (DBSCAN). Second, the authors did not in the current manuscript, but should provide control samples to demonstrate the sensitivity and dynamic range of their experimental strategy.

In STORM images displayed in Figure S1, the authors highlighted localization clusters detected by DBSCAN as a signature for integrin nanoclusters. But the authors do not discuss the localization spots that were not detected by DBSCAN. Could they be individual integrins?

And if so, they should also be considered as useful information? This brings me to another related technical question about how DBSCAN handles the case where fluorescent molecules are blinking. This is important as multiple emissions by a single fluorophore could be detected as a nanocluster of several molecules where it would be an artefact due to the photophysics of the fluorophore. Could the authors comment on these points?

Also, using isolated and stochastically physisorbed fluorophores (Ab coupled with activator/reporter pairs used in this study) on glass helped define the signature in STORM of a single isolated molecule. To obtain the signature of clustered fluorophores, the authors could use anti-donkey antibodies to cross-link those STORM-specifically labeled Ab as a means to artificially obtain clustered fluorophores. Ultimately, to avoid the bias effect of the glass surfaces on the photophysics of fluorophores and be in the same imaging conditions as for the described nanoclusters, the authors should use model systems composed of multimers of GFP vs. single GFP, immunolabeled with a GFP-binding monoclonal antibody. This will permit evaluation of the cluster signature obtained with DBSCAN analysis of STORM data for single vs. multimers of known stoichiometry. This would constitute an undisputable molecular stoichiometry ruler.

Due to the surprising finding of the nanoclusters' "universality", it is imperative for the authors to validate the findings through complementary methodologies and analytical tools. This should involve replication of results using alternative super-resolution techniques (quantitative DNA-PAINT) and exploring different clustering algorithms (Voronoi-Tessellation) to ensure the robustness and reliability of the observations.

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Author response:

As a short response to the public reviews, we would like to outline the following planned revisions:

- (1) Address the antibody concerns as indicated by reviewer 1
- (2) Assess the role of tensin (and possibly KANK), as suggested by reviewers 2 and 3, respectively.
- (3) Validate our main experimental findings using alternative super-resolution approaches, including STED to avoid potential blinking artefacts associated to standard STORM, and most possibly DNA-PAINT as a more quantitative technique, as suggested by reviewer 3.
- (4) Implement alternative analytical strategies to DBSCAN, including Voronoi tessellation as suggested by reviewer 3.
- (5) Expanded discussion on the main findings of our work and biological significance.

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